Assignment 2: Detecting Emotions with ML Algorithms

Cross-Dataset Generalization of Facial Expressions

Su Acar (2475901)
Vikram Chandrasekaran (2542544)
Henrique DaSilva Couto (2463334)

5ARE0 – Data Analysis & Learning Methods
Assignment 2 – Group 5
October 18, 2026

Example citations for later on (remove this later): (Goodfellow et al., 2013)
(Goeleven et al., 2008)
(Goeleven et al., 2008; Goodfellow et al., 2013)

1 Introduction and research questions

[Replace: Summarize the objective: facial-expression recognition on FER2013 and KDEF, evaluation across domains, explainable fuzzy modeling and practical real-time use.]

The primary question is how datasets and feature/model choices affect within-dataset accuracy and cross-dataset generalization. Secondary questions concern which emotions and representations are most transferable, whether interpretable facial geometry is useful, and how reliable a trained model remains under webcam conditions. *[Replace: Add source-supported motivation and APA citations to FER2013/KDEF after Zotero export.]*

2 Datasets, preprocessing and experimental protocol

FER2013. Images are grayscale, 48×48, with seven labels: angry, disgust, fear, happy, sad, surprise and neutral. Use the provided Training split to fit models, PublicTest for model selection/tuning, and untouched PrivateTest *only* for final evaluation.

KDEF. Use only straight/frontal (S) faces, across seven comparable expressions (KDEF "Afraid" maps to FER "fear"). Divide by identity: 80% of subjects for training, 20% for testing, with no subject overlap.

Common representation. Detect and crop KDEF faces with a margin preserving forehead and jaw; convert to grayscale, resize to 48×48 and apply the same normalization as FER. Apply identical feature extraction at training and inference. Fit any scaling, PCA or learned transformation only on training data.

Fuzzy classifier protocol. For Task 4, geometric features were extracted directly from original-resolution frontal KDEF images, rather than the FER-compatible 48×48 images. Both acquisition sessions for each identity remained in the same partition. Model selection used training subjects only, preserving the held-out KDEF subjects for final evaluation.

[Replace: Document dataset counts/class distributions; detected-face failures, deduplication/quality checks, exact train/validation/test protocol, seeds, tools, versions and potential limitations. Provide one dataset example if it supports a claim.]

3 Task 1: FER2013 baseline (20%)

Models and features. Compare Logistic Regression, SVM and Random Forest with HOG, facial landmarks and PCA features. Retain the supplied Random Forest + landmarks starter as a baseline. *[Replace: Describe representation dimensions, scaling, PCA explained variance, model configurations and how tuning on PublicTest was controlled.]*

Selection and failure analysis. *[Replace: Identify the strongest classifier/features on validation, give final PrivateTest accuracy/balanced accuracy/macro-F1, reference appendix matrices; compare per-emotion recall, class imbalance, hard-to-classify/mislabeled/lighting-varied samples.]*

Table 1: FER2013 PublicTest (validation) comparison. Fill all nine feature–model combinations; report selected model(s) separately on PrivateTest.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1
HOG	Logistic Regression	–	–	–
HOG	SVM	–	–	–
HOG	Random Forest	–	–	–
Landmarks	Logistic Regression	–	–	–
Landmarks	SVM	–	–	–
Landmarks	Random Forest	–	–	–
PCA	Logistic Regression	–	–	–
PCA	SVM	–	–	–
PCA	Random Forest	–	–	–

4 Task 2: KDEF dataset study (15%)

Use the best FER2013 feature representation and retrain the corresponding classifiers on preprocessed frontal KDEF images. Partition subjects 80/20 so each subject appears in exactly one split; document counts and random seed. *[Replace: Explain KDEF model selection without test-set leakage.]*

Table 2: KDEF held-out subject test set, with FER-selected feature representation.

Classifier	Accuracy	Bal. acc.	Macro-F1
Logistic Regression	–	–	–
SVM	–	–	–
Random Forest	–	–	–

[Replace: Explain why controlled KDEF imagery differs from in-the-wild FER2013, whether model ranking changes, the easiest/hardest emotions, and which emotions might transfer well. Cite KDEF confusion matrices in Appendix A.]

5 Task 3: Cross-dataset generalization (20%)

Test both directions required by the assignment protocol: **FER→KDEF** (FER-trained model applied to cropped/preprocessed frontal KDEF) and **KDEF→FER** (retrain the KDEF-selected configuration on its training subjects, then evaluate on FER). Keep matching 48×48 grayscale normalization and class mapping throughout; do not adapt the test-domain examples using ground-truth labels. Keep FER PrivateTest untouched until final evaluation.

Table 3: Within- and cross-dataset performance, on explicitly stated held-out sets.

Train → evaluation	Accuracy	Bal. acc.	Macro-F1
FER → FER (PrivateTest)	–	–	–
KDEF → KDEF (test subjects)	–	–	–
FER → KDEF (test subjects)	–	–	–
KDEF → FER (PrivateTest)	–	–	–

[Replace: List any model/input adaptations, define domain shift, quantify transfer degradation against in-domain comparisons, analyze per-emotion confusion, and explain which features transfer best and why. Note crop/alignment, resolution and grayscale/RGB as confounders.]

6 Task 4: Explainable fuzzy classifier (10%)

Features and protocol. We extracted seven geometric features from the 68 dlib facial landmarks: mouth width, mouth opening, left and right eye opening, eyebrow height, mouth width-to-height ratio (*smile ratio*), and mouth-corner asymmetry. Most distances were normalized by face width, while eye opening was normalized by eye width. All 980 frontal KDEF images were processed successfully, without landmark detection failures.

The predefined subject-independent split contained 56 training subjects (784 images) and 14 held-out test subjects (196 images). Within the training partition, 44 subjects (616 images) were used for fitting and 12 (168 images) for validation. MinMax scaling was fitted only on the corresponding training partition.

Fuzzy model. We trained seven one-versus-rest Takagi–Sugeno fuzzy models with PyFUME, one for each emotion. Each model uses Gaussian membership functions in its rule antecedents and a learned linear consequent. The numerical outputs are combined through fuzzy inference, and the emotion with the highest output score is selected. These scores are not calibrated probabilities.

We compared two and three clusters per emotion using macro-F1 on validation subjects. The three-cluster configuration was selected before testing and subsequently refitted on all 784 training images.

Table 4: Subject-independent KDEF fuzzy classification results.

Split	Clusters	Images	Acc. (%)	Bal. acc. (%)	Macro-F1
Validation	2	168	71.43	71.43	0.6881
Validation	3	168	73.21	73.21	0.7087
Final test	3	196	63.27	63.27	0.6082

Rule interpretation. The final system contains 21 fuzzy rules, three per emotion. A representative rule from the happiness model is:

> *Happy, Rule 1 (abridged):* IF mouth width belongs to cluster 1 (Gaussian centre 0.762) AND mouth opening belongs to cluster 1 (centre 0.204), together with the remaining geometric conditions, THEN calculate the happiness score using the fitted linear consequent.

This rule dominated 94.6% of happy training images but only 2.2% of other images. Its relatively wide-mouth configuration is consistent with smiling. The strongest rules for fear and sadness showed weaker separation, and overlapping membership functions limit their linguistic interpretation.

Evaluation and limitations. On unseen subjects, happiness (28/28), surprise (26/28) and disgust (23/28) achieved the highest recall. Fear (7/28) and sadness (8/28) were more difficult: 14 fearful faces were classified as surprise, and 12 sad faces as neutral. The roughly ten-percentage-point accuracy decrease from validation to testing indicates limited generalization across individuals. Although the fuzzy antecedents are inspectable, the linear consequents and interactions between the seven models complicate explanations of individual predictions.

7 Task 5: Creative challenge (25%)

Choose and justify **at least two** FER2013 improvements: hyperparameter tuning, feature engineering/selection, class balancing, augmentation, data curation, label verification, KDEF-derived geometric information, emotion prototypes, or domain-invariant representations. *[Replace: Name the actual two or more changes, implementation details, expected mechanism, and risks.]*

Table 5: FER2013 controlled ablation: select with PublicTest, evaluate the final choice once on PrivateTest.

Variant (PublicTest)	Accuracy	Bal. acc.	Macro-F1
Original baseline	–	–	–
Improvement A only	–	–	–
Improvement B only	–	–	–
Combined A + B	–	–	–

[Replace: Quantify gains/losses and tradeoffs. Explain class-wise and failure-case changes, computational cost, potential leakage, and one final untouched PrivateTest comparison. Reference appendix matrices.]

8 Task 6: Real-time webcam demo (10%)

Adapt the best selected model into a command-line executable Python file that opens the laptop webcam and performs live emotion recognition. Preserve the exact training face crop, alignment, feature extraction, normalization and class mapping in the inference path.

[Replace: State demo command, chosen model/.pkl, hardware/OS, measured or observed FPS/latency (if available), stability, real-world behavior, lighting/pose/occlusion issues and what was fine-tuned. Explain whether the model performed well enough, and why.]

9 Discussion, limitations and conclusion

[Replace: Synthesize: best features/classifiers on each dataset; relative difficulty of each emotion; changes of ranking; strongest and weakest transfers; fuzzy interpretability versus performance; benefits and costs of creative improvements; webcam readiness. Contrast dataset-specific and common limitations. Draw conclusions only from actual measurements and concrete error examples.]

Declaration of AI tool use

During the preparation of this work, I/We used [NAME TOOL / SERVICE / VERSION OF AI TOOL] in order to [REASON]. The following parts of the assignment were affected/generated by AI tool usage: [INTRODUCTION / METHODS / xxx, DISCUSSION]. After using this tool/service, [NAME STUDENT(S)] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [NAME STUDENT(S)] take(s) full responsibility for the content of their work.

References

Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska Directed Emotional Faces: A validation study. *Cognition & Emotion*, *22*(6), 1094–1118. https://doi.org/10.1080/02699930701626582

Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In M. Lee, A. Hirose, Z.-G. Hou, R. M. Kil, D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, & G. Weikum (Eds.), *Neural Information Processing* (pp. 117–124, Vol. 8228). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-42051-1_16

A Confusion matrices for reported models

[Replace: Replace every relevant slot with an actual confusion matrix exported by the corresponding notebook. Remove unreported model configurations; include matrices for all reported final models, with evaluation split, class ordering, count or normalization labeled.]

A.1 FER2013 baseline models (PublicTest; final chosen model on PrivateTest)

HOG + Logistic Regression

cm_fer_hog_logreg.pdf

HOG + SVM

cm_fer_hog_svm.pdf

HOG + Random Forest

cm_fer_hog_rf.pdf

Landmarks + Logistic Regression

cm_fer_lm_logreg.pdf

Landmarks + SVM

cm_fer_lm_svm.pdf

Landmarks + Random Forest

cm_fer_lm_rf.pdf

PCA + Logistic Regression

cm_fer_pca_logreg.pdf

PCA + SVM

cm_fer_pca_svm.pdf

PCA + Random Forest

cm_fer_pca_rf.pdf

FER final selected (PrivateTest)

cm_fer_final_private.pdf

A.2 KDEF, cross-dataset, fuzzy and creative models

KDEF Logistic Regression

cm_kdef_logreg.pdf

KDEF SVM

cm_kdef_svm.pdf

KDEF Random Forest

cm_kdef_rf.pdf

FER → KDEF

cm_fer_to_kdef.pdf

KDEF → FER

cm_kdef_to_fer.pdf

Creative variant(s)

cm_creative_final.pdf

Additional reported model

cm_additional.pdf

A.3 Fuzzy classifier: held-out KDEF subjects

Table 6: Final fuzzy classifier confusion matrix (counts; 196 test images from 14 unseen subjects). Rows are true labels and columns are predictions.

True / Pred.	Af	An	Di	Ha	Ne	Sa	Su
Afraid	7	1	2	0	2	2	14
Angry	0	14	10	0	2	2	0
Disgust	0	3	23	1	0	1	0
Happy	0	0	0	28	0	0	0
Neutral	0	3	0	0	18	6	1
Sad	0	6	2	0	12	8	0
Surprise	0	0	0	0	1	1	26

Af: afraid; An: angry; Di: disgust; Ha: happy; Ne: neutral; Sa: sad; Su: surprise.